MaskGeo: Masked Learning for Cross-view Geo-localization

Anonymous CVPR submission

Abstract

Existing cross-view geo-localization (CVGL) methods have primarily focused on learning discriminative features without considering the impact of limited field of view (FoV), orientation, and visible content overlap between ground and aerial images. In this work, we propose a novel masked learning strategy to address these challenges in CVGL. Our training pipeline masks ground and aerial images to introduce semi-positive ground and aerial image pairs. Prior methods avoid aerial image augmentations that introduce blank pixels, due to the destructive nature of this signal; we show that this destructive signal can be overcome by utilizing prior knowledge of the FoV overlap between ground and aerial images. We achieve this by a masked learning strategy which enables the model to discriminate between positives and negatives as well as rank positives and semi-positive image pairs. We adopt Ranking InfoNCE (RINCE) to enforce that positives rank above semi-positives, which rank above negatives. Experiments on existing benchmark CVGL datasets demonstrate that our method performs competitively with the state-of-the-art methods while also being robust to FoV and orientation variations.

1. Introduction

The process of estimating the geographic location of a query ground level image using a reference image from a different perspective is termed cross-view geo-localization (CVGL). CVGL is applicable to a broad range of tasks such as digital forensics and disaster situational awareness [11, 35], autonomous navigation and augmented reality in GPS-denied environments [29]. With iterative advancements, CVGL methods have saturated existing benchmark datasets such as CVUSA and CVACT [5, 37, 40]; however, CVGL still remains challenging in practical scenarios.

CVGL has been framed as an image retrieval task since its introduction [13]. Due to the powerful feature extraction capabilities of deep convolution network based methods and the advancement of deep metric learning techniques on tasks such as face recognition, earlier CVGL methods adopted Siamese networks and metric learning based methods [14].

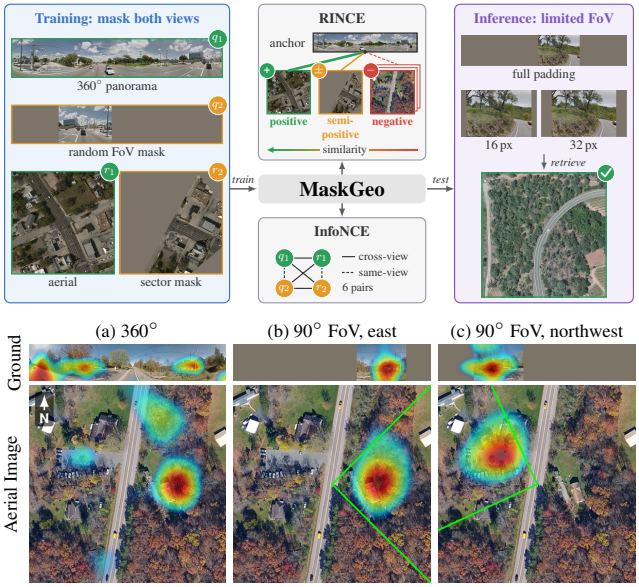


Figure 1. Top: overview of MaskGeo. In training, a 360° panorama (q_1) and its aerial image (r_1) are paired with a randomly FoV-masked panorama (q_2) and a sector-masked aerial image (r_2). InfoNCE is applied to all six pairs of the four views, and RINCE ranks, in both directions, the full image of the other domain (positive) above its masked version (semi-positive) and above images of other locations (negatives). At inference, a limited-FoV query retrieves the matching aerial image. Bottom: ground and aerial attention for a full 360° panorama query (a) and for 90° FoV queries cut from the same panorama facing east (b) and north west (c). The green sector marks the visible part of the ground image crop.

Until recently, CVGL methods assumed a known field of view (FoV) and orientation; this assumption limits their practical applicability. To be robust to orientation and FoV variations, DSM [24] estimates the orientation difference of the ground image in the feature space and shifts and crops the aerial image feature map accordingly. ArcGeo [26] achieved higher evaluation metrics for multiple FoVs, although the model was trained at a 90° FoV; it also demonstrated the effects of hard sampling and batch size for batch-all losses. Several other works push evaluation metrics, but they were designed for specific FoVs. ConGeo [18] optimizes for FoV-agnostic retrieval, extending Sample4Geo [5], and proves to be more robust to FoV and

orientation variations. Subsequently, FoV-robust methods such as SinGeo [2] and HCLGeo [3] attribute the lower performance of existing methods at narrower FoVs to the learning paradigm, and propose curriculum learning and distillation learning, respectively. SinGeo's curriculum learning strategy further enhances the FoV and orientation robustness of ConGeo by gradually introducing harder CVGL pairs as training progresses. HCLGeo's distillation learning trains a teacher model for panoramic images and then distills the knowledge into student models trained on lower FoVs; finally, a mixture-of-experts consolidation mechanism produces the final retrieval vector.

We hypothesize that contrastive learning becomes destructive under masking because it forces the embedding of a limited-FoV or masked view to match that of the full view, even though the two share only part of their content. This may also explain why prior work avoids aerial masking and continuous rotation [2]. We show that prior knowledge of orientation and FoV, available at training time, can overcome this destructive signal. Masking both views yields semi-positives: views of the correct location that share only part of their content with the anchor. We weight each InfoNCE term by the visible overlap between its views and, rather than forcing a masked view to be either a positive or a negative, we rank it between the two with RINCE [8]. The result is a single model that is more robust to FoV and orientation variations.

2. Related Work

2.1. Discriminative Learning in Cross-view Geo-localization

CVGL is framed as a retrieval problem where ground-view imagery is matched against a set of geotagged aerial imagery to determine its geolocation [13]. In contrast to single-view image retrieval, images from different domains need to be mapped to the same embedding space. Initially, CVGL was conducted using handcrafted features [13], where ground image features were used to produce a probability function over a geographic area. Workman et al. [32] trained an aerial image encoder to minimize the Euclidean distance between the aerial image features and ground image features extracted from an image encoder pretrained on the Places dataset [38]. The max-margin triplet loss was heavily used in retrieval tasks; however, the margin needs to be carefully calibrated. Vo and Hays [28] explored a soft-margin triplet loss for discriminative learning, alleviating the need for a fixed margin, and proposed a batch-all triplet loss. The authors of CVM-Net [9] utilize separate local feature extractors for the two image domains and explore the use of shared global feature extractors; they utilize a weighted soft-margin triplet loss to improve convergence speed. TransGeo [40] proposed a pure transformer-based method using the triplet loss for discriminative learning. To retain the spatial information of the local features, some methods avoid global pooling layers and utilize optimal transport [25] or attention mechanisms [12, 31] on the feature maps. In contrast to triplet losses, which need to be computed for all triplets within a batch for improved discriminative learning, the ArcGeo loss [26] is a novel batch-all loss function based on cross entropy. The ArcGeo loss was inspired by loss functions used in face recognition [4], a task which resembles CVGL. Similarly, Sample4Geo [5] utilizes InfoNCE [19], which contrasts an anchor with all other samples in the batch, resulting in effective discriminative learning. Moreover, Sample4Geo was one of the first works to show that weight-shared feature extractors have better CVGL performance. InfoNCE has been used in the most recent work to date [2, 3], and the discriminative learning of batch-all loss functions is enhanced by hard negative sampling [5].

2.2. Limited FoV Cross-view Geo-localization

Limited-FoV CVGL has been actively researched in recent years [2, 3, 6, 12, 18, 26, 27]. The classical CVGL benchmarks of north-aligned panoramas and the corresponding aerial image tiles [15, 32] are saturated by the advances in CVGL techniques. However, such benchmark datasets do not reflect most practical scenarios with limited FoV and unknown orientation. Works focusing on limited-FoV settings crop and roll ground image panoramas along the azimuth direction from standard benchmarks to resemble unknown-orientation and limited-FoV imagery [2, 3, 18, 26].

To reduce the perspective differences, researchers use polar transforms on the aerial image to create pseudo ground imagery [23], and some methods create bird's-eye-view images from panoramic ground imagery [30, 34]. DSM [24] jointly discriminates and computes the correlation between features of limited-FoV ground images and the polar-transformed aerial image along the azimuth angle axis to predict the most feasible orientation. Rodrigues et al. [22] improve DSM results by incorporating partially visible global features for enhanced limited-FoV CVGL, and are among the first to consider aerial image sectoring: during training, a FoV crop of a ground image, the corresponding aerial image sector and the full aerial image are given. ArcGeo [26] introduces a novel loss function and data augmentation techniques to train a single model for FoV-robust CVGL. Similarly, ConGeo [18] works towards a robust architecture inspired by Sample4Geo [5] that can be trained once and evaluated on different FoVs. Its image encoders aim to minimize the feature distance between ground image panoramas, FoV crops, aerial images and augmented aerial images from the same location jointly using the InfoNCE loss function. SinGeo [2] adopts the architecture of Con-

Geo, introduces a curriculum learning paradigm and produces state-of-the-art results on the CVACT dataset. SinGeo uses fixed quarter rotations of aerial images, since continuous rotations lose information at the edges and introduce blank pixels. HCLGeo [3] utilizes a continual learning and mixture-of-experts paradigm for FoV-robust CVGL, where a teacher model is trained on panoramic CVGL and student models are trained continually on limited-FoV images. Similar to SinGeo and ConGeo, HCLGeo retains panoramic image supervision through the teacher model; in addition, HCLGeo uses expert modules for multiple tasks, which also excel on unseen FoVs.

2.3. Data Augmentation in CVGL

Data augmentation is used in CVGL to tackle overfitting and improve generalization. However, only a limited set of data augmentations that retain spatial information are applied [37]. On the other hand, image transformations such as polar transforms are applied to bridge the large perspective change between aerial and ground imagery [23, 34]. Vaclnet [30] uses satellite image masking corresponding to the orientation and FoV of the limited-FoV ground image to encourage content-based retrieval during training and global image understanding. Rodrigues et al. [21] created augmented ground imagery by masking objects from ground-level images to address the temporal variation between ground and aerial images, where the original and augmented samples were used at train time. To improve generalization, GeoDTR [37] used synchronous aerial and ground image rotation, which retains low-level image features and geometric layout consistency between aerial and ground images. ArcGeo [26] uses FoV cropping on panoramic images. In addition, standard image augmentations such as color jitter, dropout, synchronous horizontal flipping and rotation of aerial and ground images are used throughout the literature [2, 5, 18, 26, 33].

3. Methodology

To further increase single-model robustness for limited-FoV images, this paper introduces a novel training strategy that guides training using pseudo labels at train time. This technique works effectively for both CNN- and ViT-based image encoders. This section illustrates how we can effectively utilize the discriminative power of contrastive learning and learn a better representation using the semi-positive samples from the same location.

3.1. Background to CVGL

Modern CVGL methods are based on deep metric learning. Here an image encoder is trained to contrast between correct and incorrect aerial images given a ground-level query image. A cosine similarity matrix is used for optimization, where the diagonal elements represent the similarity of positive pairs from the same location. During training, positive pairs are pulled closer and negative pairs are pushed apart. Recent CVGL techniques heavily utilize the InfoNCE contrastive loss, since its formulation uses all batched samples for contrastive learning: each positive is compared against $N-1$ negatives, where N is the batch size. The performance of contrastive CVGL methods is greatly enhanced by batching hard negative samples during training, a sampling strategy commonly used in CVGL [5]. Moreover, CVGL methods apply InfoNCE in both directions to enable information flow both ways, following [20].

Given a query image, the InfoNCE objective contrasts its positive against a set of reference image samples, as shown in Eq. (1):

$$\mathcal{L}_{\mathrm{NCE}}(q, R) = -\log \frac{e^{s(q,r_+)/\tau}}{\sum_{r_i \in R} e^{s(q,r_i)/\tau}}, \qquad (1)$$

where q denotes the query image embedding, R is a batch of reference image embeddings r_i, $r_+ \in R$ is the positive reference for the query, $s(\cdot,\cdot)$ is the cosine similarity and τ is a learnable temperature parameter.

Following SinGeo [2] and ConGeo [18], we adopt a dual discriminative learning architecture that aims for a single model robust to multiple FoVs (Fig. 2), together with hard negative sampling based on Sample4Geo [5].

3.2. Curriculum Masked Learning

Prior CVGL methods restrict aerial augmentation to pixel-level changes and 90-degree rotations, since masking and continuous rotation discard information [2, 37]. We argue that this can be beneficial when it is controlled. A ground image with a limited FoV observes only a sector of its aerial tile, so masking the aerial image to a sector mirrors the limited-FoV setting on the aerial image. This extends the retrieval of positive references to both positive and semi-positive pairs. MaskGeo learns from ranking semi-positive references (same location, partial overlap) to further enhance the learnt representation. Since heavily masked pairs are hard to match early in training, we introduce masking gradually through a curriculum [1]. SinGeo [2] applies such a curriculum to the ground FoV; we adopt its schedule and extend the curriculum to the aerial view (Fig. 2). Each location in a batch therefore contributes four views: the ground panorama q_1, its FoV-masked crop q_2, the aerial tile r_1 and its sector-masked counterpart r_2. We use the same symbols for their embeddings, and write Q_k and R_k for the batch of ground and aerial embeddings of view k.

Ground masking. The ground panorama is rolled by a random azimuth and cropped to a field of view FoV centered on heading ψ. The crop is then padded to a fixed width; we compare padding strategies in the ablation study (Sec. 4.3).

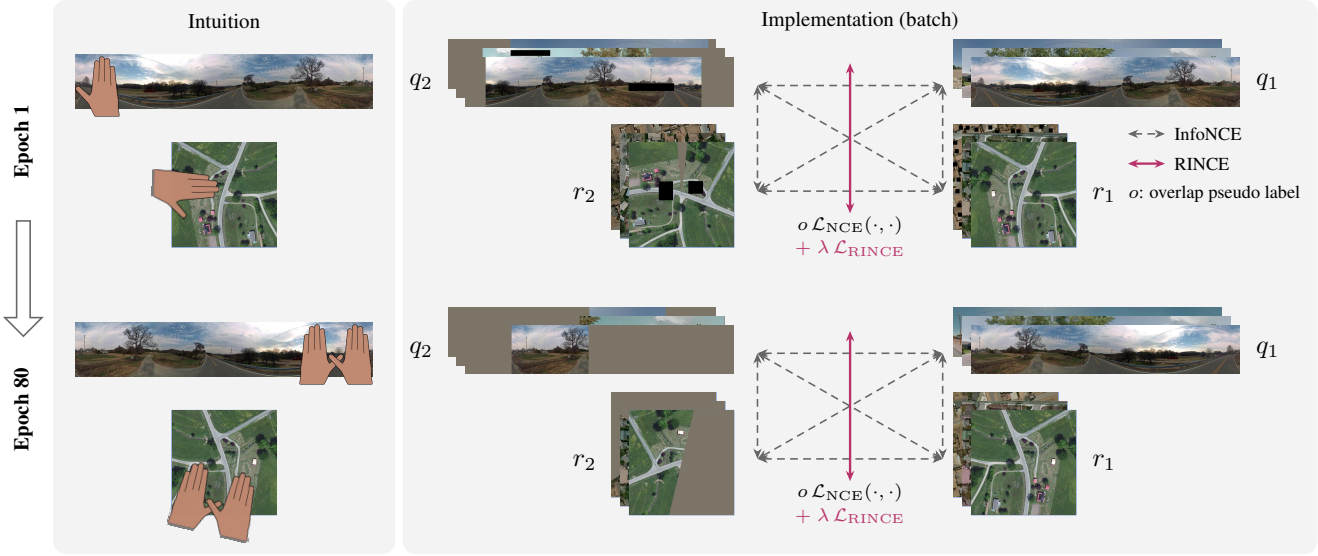


Figure 2. The masked learning task. Left: intuition. Hands hide the part of the original ground panorama q_1 and aerial image r_1 that is masked out, so the pair must be matched from the content that remains visible. Right: implementation on a training batch. The masked views q_2 (cropped and padded) and r_2 (wedge-masked) are contrasted with the originals q_1 and r_1 using InfoNCE between every pair of images, weighted by the overlap of their visible content. RINCE is applied only across domains, in both directions: each ground view ranks r_1 above r_2 above the hardest aerial negative, and each aerial view ranks q_1 above q_2 above the hardest ground negative. As training progresses from epoch 1 to epoch 80, the visible ground FoV narrows and the masked aerial sector widens.

Aerial masking. The aerial image is masked to a sector with center c and angular extent w, both in degrees. The sector is anchored to the ground crop: its center is the ground heading offset by a deviation δ,

$$c = (\psi + \delta) \bmod 360, \tag{2}$$

therefore δ controls how far the visible aerial sector points away from what the ground image sees. The masked aerial image is then rotated counter-clockwise by a continuous angle ρ. Unlike δ, the rotation ρ does not decide which part of the scene is visible; it is an image augmentation that breaks the north alignment of the pair.

Curriculum. Let $t \in [0, 1]$ denote training progress. Following SinGeo [2], the ground FoV shrinks from 360° to $\mathrm{FoV}_{\min}$, quickly at first and then more slowly,

$$\begin{aligned} \mathrm{FoV}(t) &= 360 - (360 - \mathrm{FoV}_{\min})\, f(t), \\ f(t) &= \frac{1 - \exp(-\gamma t)}{1 - \exp(-\gamma)}, \end{aligned} \tag{3}$$

where $\gamma > 0$ controls the decay rate and $\mathrm{FoV}_{\min} = 70°$. The aerial sector extent shrinks linearly from 360° to 180°,

$$w(t) = 360 - 180\, t, \tag{4}$$

and the deviation and rotation are sampled independently from a symmetric range that widens linearly,

$$\delta, \rho \sim \mathcal{U}(-\theta_{\max}, \theta_{\max}), \qquad \theta_{\max}(t) = 180\, t. \tag{5}$$

At the start of training ($t = 0$) both images are unmasked and aligned, so the pair overlaps fully. By the end ($t = 1$) the ground view covers 70°, the aerial sector covers 180°, and the sector can face any direction relative to the ground crop, so overlap is no longer guaranteed. The overlap between the two masks gates the contrastive loss described in Sec. 3.3.

Pseudo labels. For any two views x and y of the same location, the curriculum yields an overlap pseudo label $o(x, y) \in [0, 1]$: the angular overlap of the azimuth ranges visible in x and y, divided by 360°. For example, $o(q_1, r_1) = 1$ and $o(q_2, r_1) = \mathrm{FoV}/360°$. The pseudo label gates the contrastive loss, so that pairs with low overlap contribute less. This allows the model to focus on learning from pairs with high overlap, which are more informative for the task.

3.3. Gated InfoNCE Loss

Prior work in CVGL has shown that the InfoNCE loss is effective, yet it is sensitive to the destructive signal that arises from enforcing similarity between a query and a masked reference image. To control this signal, we weight each InfoNCE term by the overlap pseudo label of the anchor and its positive,

$$\mathcal{L}^{\mathrm{g}}_{\mathrm{NCE}}(q, R) = o(q, r_+)\, \mathcal{L}_{\mathrm{NCE}}(q, R). \tag{6}$$

This gates the contrastive signal of each pair by how much content its two views share. It also removes the need

for hand-set loss weights between same-domain and cross-domain pairs used in recent CVGL methods [2, 18]. The gated loss is applied to all six pairs of the four views, four cross-domain and two same-domain,

$$\mathcal{L}^{\text{tot}}_{\text{NCE}} = \sum_{(x,Y)\in\mathcal{P}} \mathcal{L}^{\text{g}}_{\text{NCE}}(x, Y),$$
$$\mathcal{P} = \{(q_1, R_1), (q_2, R_1), (q_1, R_2), (q_2, R_2), (q_1, Q_2), (r_1, R_2)\}, \tag{7}$$

where each term is averaged over the anchors x in the batch.

3.4. Ranking Positives and Semi-positives with RINCE

Masking turns the binary positive/negative split of InfoNCE into three tiers, in both retrieval directions. For a ground anchor $q \in \{q_1, q_2\}$, the full aerial tile r_1 of the same location is a *positive*, its sector-masked counterpart r_2 is a *semi-positive*, and aerial images of other locations are *negatives*. Symmetrically, for an aerial anchor $r \in \{r_1, r_2\}$, the panorama q_1 is a positive, the FoV-masked crop q_2 is a semi-positive, and ground images of other locations are negatives. A binary contrastive objective must place each masked view in one of two sets. Treating it as a positive forces the embedding of a partial view onto that of the full view; on the ground side this collapses a limited-FoV crop onto its panorama, which is the destructive signal discussed in Sec. 3.3. Treating it as a negative instead discards the fact that it depicts the same location. Neither reflects that a masked view contains a subset of the content of its unmasked counterpart.

We therefore adopt Ranking InfoNCE (RINCE) [8], which replaces the binary split with ranked sets of positives. For an anchor a with positive p, semi-positive $\tilde{p}$ and negative n from the other domain, we enforce

$$s(a, p) \;>\; s(a, \tilde{p}) \;>\; s(a, n), \tag{8}$$

where $s(\cdot,\cdot)$ is the cosine similarity of Eq. (1). For ground anchors $(p, \tilde{p}) = (r_1, r_2)$; for aerial anchors $(p, \tilde{p}) = (q_1, q_2)$. This ordering requires no annotation. Both masked views are obtained by removing content from their unmasked counterparts, so the overlap of any view of the other domain with the full view is never smaller than its overlap with the masked view. The ranks are thus given by construction, analogous to the frame-, shot- and video-level positives of [8].

RINCE applies InfoNCE recursively. The rank-1 term treats p as the positive and every lower-ranked sample, including the semi-positive, as a negative; the rank-2 term drops p and treats $\tilde{p}$ as the positive. As each rank contains a single positive, we use the RINCE-uni variant:

$$\ell_1(a) = -\log \frac{e^{s(a,p)/\tau_1}}{e^{s(a,p)/\tau_1} + e^{s(a,\tilde{p})/\tau_1} + e^{s(a,n)/\tau_1}}, \tag{9}$$

$$\ell_2(a) = -\log \frac{e^{s(a,\tilde{p})/\tau_2}}{e^{s(a,\tilde{p})/\tau_2} + e^{s(a,n)/\tau_2}}, \tag{10}$$

$$\mathcal{L}_{\text{RINCE}} = \frac{1}{|\mathcal{A}|} \sum_{a\in\mathcal{A}} \big(\ell_1(a) + \ell_2(a)\big), \tag{11}$$

where $\mathcal{A}$ contains the ground and aerial anchors of every location in the batch. The two terms compete: ℓ_1 pushes the semi-positive away from the anchor while ℓ_2 pulls it closer than the negative, so the semi-positive settles between the positive and the negatives instead of collapsing onto either. In the aerial-anchored direction, the rank-1 term requires $s(r, q_1) > s(r, q_2)$, which directly prevents the embedding of a limited-FoV crop from collapsing onto that of its panorama. We apply $\mathcal{L}_{\text{RINCE}}$ across domains only, and use $\tau_1 = \tau_2 = 0.5$.

Negatives. RINCE ranks positives against a set of negatives at the lowest rank. Since the InfoNCE terms in Eq. (7) already contrast every anchor against all other locations in the batch, we instantiate this set with a single hard negative,

$$n = \arg\max_{v\in\mathcal{V}^-(a)} s(a, v), \tag{12}$$

where $\mathcal{V}^-(a)$ contains the full and masked views of the other domain from all other locations in the batch. We do not rank negatives among themselves: in preliminary experiments, ranking them by geographic distance was ineffective, as image similarity does not correlate with geographic distance.

Overlap. Unlike the InfoNCE terms, $\mathcal{L}_{\text{RINCE}}$ is not weighted by the overlap pseudo label o. A semi-positive with little or no visible overlap with the anchor is still ranked above negatives. This encodes the prior that a view of the anchor's own location is more similar to it than a view of a different location, while ℓ_1 keeps it below the full view.

Relation to Rank-N-Contrast. Rank-N-Contrast (RnC) [36] extends this construction to continuous labels by ranking every batch sample by label distance to each anchor. For our three discrete tiers, the two objectives coincide up to normalization and temperature, and we adopt RINCE because our ranks are given by construction. Ranking several masked views by continuous overlap, where RnC applies naturally, is left for future work.

The final training objective is

$$\mathcal{L} = \mathcal{L}^{\text{tot}}_{\text{NCE}} + \lambda\, \mathcal{L}_{\text{RINCE}}, \tag{13}$$

where $\lambda = 0.25$.

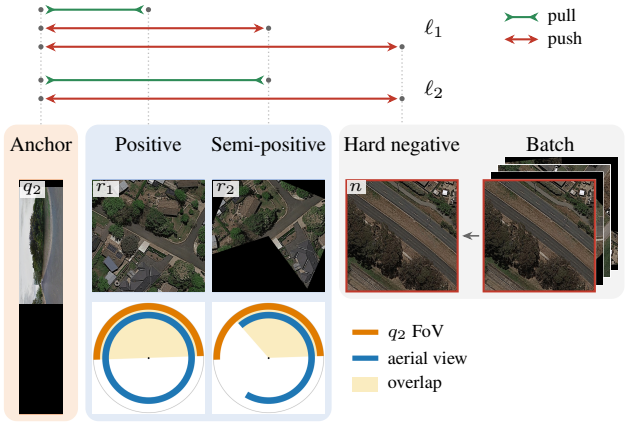


Figure 3. RINCE ranking for a masked ground anchor q_2. Columns, in rank order from left to right: the full aerial tile r_1 (positive), its sector-masked counterpart r_2 (semi-positive), and the hardest negative n (red). n is picked from the tiles and sectors of the other locations in the batch, which are stacked together. Below r_1 and r_2 is each one's azimuth overlap with the FoV of q_2. Because r_2 is masked from r_1, its overlap can never exceed that of r_1, which fixes the ranking. Arrows show the forces of the two RINCE terms: ℓ_1 pulls r_1 towards q_2 and pushes r_2 and n away; ℓ_2 pulls r_2 towards q_2 and pushes n away. Aerial anchors are ranked symmetrically: $q_1 > q_2 >$ the hardest ground negative.

4. Experiments

4.1. Experimental Setup

Datasets CVUSA and CVACT are two standard benchmark datasets for CVGL with north-aligned ground and aerial image pairs. Each contains 35,532 training pairs and 8,884 validation pairs. In addition, CVACT contains 92,802 testing image pairs. Both datasets contain panoramic ground images with a FoV of 360° and their aerial images. The VIGOR [39] dataset is a more challenging dataset for CVGL where a query image has an uncentered positive reference and multiple semi-positive reference images. In our experiments we will only consider the positive reference image for evaluation.

Metrics We use standard retrieval metrics for CVGL. Recall at K (R@K) is the percentage of queries whose correct reference is among the top K retrieved references; R@1% uses the top 1% of the reference set. We report R@1, R@5, R@10 and R@1% for both datasets. This evaluation tests the cosine similarity based retrieval accuracy of the model.

Implementation Details We use a ConvNeXt-B [16] backbone for the image encoder. We train the model for 80 epochs with a batch size of 16 using AdamW [17] with a cosine scheduled learning rate starting at 1e-4.

4.2. Comparison with State-of-the-art Methods

CVUSA & CVACT We compare MaskGeo with the state-of-the-art CVGL methods on two standard CVGL datasets. We use the CVUSA and CVACT datasets for this work. MaskGeo is compared against prior state-of-the-art methods [2, 3, 5, 7, 9, 18, 22, 24, 25, 41] under unknown orientation and limited-FoV settings. Methods such as CongGeo, SinGeo and HCLGeo require only one training run for multiple FoVs. MaskGeo achieves competitive performance using a simpler learning paradigm and architecture than HCLGeo. On CVACT in particular, MaskGeo obtains significantly higher results for non panoromic imagery. We further evaluate our method using 4px, 32px and 64px fixed padding at evaluation time, to demonstrate that MaskGeo is agnostic to the padding amount. We report the results in Tab. 1.

VIGOR is a more challenging dataset for CVGL where the query image has an uncentered positive reference image. MaskGeo is compared against prior state-of-the-art methods [2, 3, 5, 18] under unknown orientation and different FoV settings. MaskGeo achieves significant performance gains due to masked learning and the FoV overlap prior between ground images and aerial image sectors. The results are reported in Tab. 2.

4.3. Ablation Study

We show the contribution of each component introduced in this work, starting from the SinGeo baseline. All experiments are done on the CVUSA dataset. On top of SinGeo, we first add curriculum-based augmentation of the ground and aerial imagery. Introducing aerial image masking reduces the accuracy significantly. This is because of the information loss and the enforcement of similarity between full ground images and partial aerial images. We believe that prior methods overcome this by lowering the InfoNCE weights between augmented pairs.

To overcome this, we utilize gated InfoNCE, removing the need for hand-set InfoNCE weights. Gated InfoNCE alone performs poorly during the later stage of training, where the FoV overlap between the two views is reduced. Contrastive learning can be applied without a destructive signal. However, InfoNCE alone cannot express partial similarity; we therefore add the RINCE ranking term, which places each semi-positive between its positive and the hardest negative. Table 3 lists the contribution of each step in order.

4.4. Padding Agnostic - Masked Ground Imagery

Our experiments demonstrate that MaskGeo is agnostic to the amount of horizontal padding, making it practical for CVGL scenarios with unknown FoVs. Figure 4 presents the R@1 performance of MaskGeo across different padding widths for FoVs of 70°, 90°, 180°, 270° and 360°. This

Table 1. Comparison with state-of-the-art methods on CVUSA and CVACT under unknown orientation and limited FoV. Avg is the mean R@1 over the four FoVs. The MaskGeo (16px), (32px) and (64px) rows use a fixed padding of 16px, 32px and 64px at test time, highlighting the robustness to the padding amount.

	Method	Avg R@1	FoV 360°				FoV 180°				FoV 90°				FoV 70°			
			R@1	R@5	R@10	R@1%	R@1	R@5	R@10	R@1%	R@1	R@5	R@10	R@1%	R@1	R@5	R@10	R@1%
CVUSA	DSM [24]	37.9	78.1	89.5	92.9	98.5	48.5	68.5	75.6	93.0	16.2	31.4	39.9	71.1	8.8	19.9	27.3	61.2
	SEH [7]	40.9	85.4	93.5	95.8	–	53.7	72.3	79.0	–	16.6	32.2	40.3	–	7.8	18.8	25.6	–
	GAL [22]	–	–	–	–	–	48.9	69.9	78.5	95.7	22.5	44.4	54.2	83.6	15.2	32.9	42.1	75.2
	Sample4Geo [5]	68.5	93.3	97.5	98.0	99.1	84.6	95.9	97.6	99.5	55.1	78.3	85.0	96.6	40.9	65.4	74.1	93.0
	ConGeo [18]	73.4	96.6	98.9	99.2	99.7	92.3	97.9	98.7	99.7	55.5	75.4	81.5	75.4	49.1	70.8	78.0	93.1
	SinGeo [2]	79.1	96.8	99.0	99.2	99.5	91.8	97.6	98.4	99.5	70.1	86.6	91.0	97.5	58.0	78.1	83.9	95.2
	HCLGeo [3]	**83.3**	**97.6**	99.1	–	–	**92.4**	97.4	–	–	75.2	89.5	–	–	**68.3**	85.5	–	–
	MaskGeo (Ours)	82.5	94.6	97.6	98.1	99.1	92.1	97.6	98.5	99.6	**76.2**	90.0	93.4	98.3	66.9	84.7	89.3	97.1
	MaskGeo (16px)	80.6	96.1	98.7	99.0	99.7	91.7	97.3	98.3	99.6	74.9	89.4	92.5	97.9	59.7	78.9	84.2	94.7
	MaskGeo (32px)	81.8	96.1	98.9	99.2	99.8	91.8	97.4	98.5	99.6	75.7	90.0	93.1	98.0	63.6	81.7	86.8	96.0
	MaskGeo (64px)	**82.6**	96.1	98.8	99.3	99.7	91.9	97.4	98.4	99.6	76.0	90.3	93.4	98.2	66.4	84.2	88.9	96.9
CVACT	DSM [24]	37.1	72.9	85.7	88.9	95.3	49.1	67.8	74.2	89.9	18.1	33.3	40.9	68.7	8.3	20.7	27.1	57.1
	SEH [7]	36.5	77.4	88.6	90.9	–	47.7	67.9	74.3	–	13.9	28.4	36.2	–	6.9	16.5	22.3	–
	GAL [22]	–	–	–	–	–	49.9	68.5	77.2	93.0	26.1	49.2	59.3	85.6	14.2	33.0	43.2	77.2
	Sample4Geo [5]	47.0	82.4	90.6	92.3	95.4	58.9	79.8	85.3	95.0	27.9	52.0	62.3	87.0	18.8	40.4	51.0	81.3
	ConGeo [18]	54.6	83.0	90.6	92.4	96.3	**70.3**	85.2	88.6	95.1	40.6	62.6	69.8	86.6	24.6	45.3	54.3	80.6
	SinGeo [2]	55.8	**83.4**	91.1	92.6	95.7	68.0	82.4	85.8	93.0	42.6	65.5	72.8	89.0	29.0	51.8	61.0	83.4
	HCLGeo [3]	55.4	80.3	88.3	–	–	66.4	79.7	–	–	40.6	60.9	–	–	34.3	55.6	–	–
	MaskGeo (Ours)	**56.4**	72.3	81.2	83.9	89.2	69.3	83.6	87.3	94.9	**46.7**	67.1	74.0	89.2	**37.3**	59.1	66.8	86.2
	MaskGeo (16px)	56.9	82.1	90.0	92.0	95.9	69.6	83.7	87.6	94.8	44.3	66.14	73.0	88.5	31.7	53.8	62.2	83.6
	MaskGeo (32px)	57.9	82.2	90.0	92.0	96.0	69.6	83.9	87.6	94.8	45.1	66.7	73.4	88,8	34.5	56.8	65.0	85.1
	MaskGeo (64px)	**58.3**	82.1	90.1	91.8	96.0	69.3	83.6	87.5	94.9	45.4	67.3	73.7	89.0	36.4	58.4	66.6	86.0

Table 2. Comparison with state-of-the-art methods on the cross-area and same-area splits of VIGOR under unknown orientation and limited FoV.

Method	Cross-Area				Same-Area			
	FoV 360°		FoV 90°		FoV 360°		FoV 90°	
	R@1	R@1%	R@1	R@1%	R@1	R@1%	R@1	R@1%
ConGeo [18]	16.2	72.9	3.9	54.3	61.9	98.4	8.5	68.7
SinGeo [2]	24.7	85.7	4.9	63.7	62.9	98.4	24.0	91.5
HCLGeo [3]	29.8	**85.8**	4.7	58.2	69.3	97.8	30.4	93.0
MaskGeo (Ours)	**31.0**	85.5	**7.2**	**70.0**	–	–	–	–

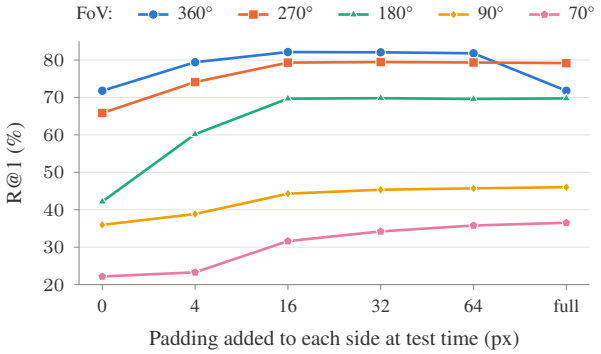


Figure 4. R@1 of MaskGeo on the CVACT validation set for different amounts of blank padding added to each side of the ground image at test time. At 0 the crop is not padded; *full* pads it to the width of the 360° panorama. For the 360° panorama, *full* adds no padding and equals 0.

insensitivity to padding width also suggests that MaskGeo can be readily adopted by ViT-based architectures without requiring architectural modifications.

In this work, masking the ground image panorama effectively introduces horizontal padding, placing the ground image crop within a larger canvas. Our ablation study demonstrates that the image encoder exploits this padding, suggesting that it helps alleviate the blind spots introduced by convolutional layers [10]. By introducing horizontal padding, MaskGeo preserves the useful information contained in the edge pixels of the ground image, mitigating this effect. This becomes particularly important when considering narrower FoV crops, where the contribution of edge pixels is more significant. The effect of vertical padding is further investigated in the supplementary material.

4.5. Heatmap Activation Analysis

We show the results of enforcing FoV overlap during training through activation heatmaps on MaskGeo. We use an input that is padded by a fixed amount of 64px and compare our results with SinGeo trained on a similarly padded ground image branch. MaskGeo aerial activations are more orientation- and FoV-informed. We quantify this through

Table 3. Ablation study on CVUSA. Each row adds one component to the row above, starting from the SinGeo baseline. Avg is the mean R@1 over the four FoVs.

Component added	FoV 360°		FoV 180°		FoV 90°		FoV 70°		Avg R@1
	R@1	R@1%	R@1	R@1%	R@1	R@1%	R@1	R@1%	
SinGeo (baseline)	–	–	–	–	–	–	–	–	79.1
+ Aerial image masking	90.5	–	89.6	–	68.3	–	55.0	–	–
+ Ground image padding	–	–	–	–	–	–	–	–	79.1
- InfoNCE weights	93.3	–	83.1	–	57.6	–	46.1	–	70.0
+ Gated InfoNCE	92.4	98.5	91.0	99.4	72.9	97.9	62.5	96.3	–
+ MaskGeo curriculum schedule	–	–	–	–	–	–	–	–	79.1
+ RINCE loss (Ours)	–	–	–	–	–	–	–	–	–

Table 4. Mean fraction of aerial activation mass inside the visible sector, h (%), for ground queries with 64px padding. Higher is better.

Method	270°	180°	90°	70°
SinGeo [2]	85.1	71.0	51.5	45.4
MaskGeo (Ours)	**91.5**	**80.5**	**59.8**	**52.0**

a metric that measures the fraction of activation mass that falls within the visible region of the aerial image, $h = m_{\text{in}}/m_{\text{all}}$, where m_{in} is the activation mass inside the visible sector and m_{all} is the total activation mass. A lower h indicates that the heatmap is scattered over the aerial image and a higher h indicates that the heatmap is concentrated within the visible aerial sector. Figure 5 compares the aerial heatmaps of SinGeo and MaskGeo for limited-FoV ground queries. SinGeo activates regions outside the visible sector, whereas MaskGeo concentrates its activations within it. Table 4 reports the mean h for both methods across four FoV settings.

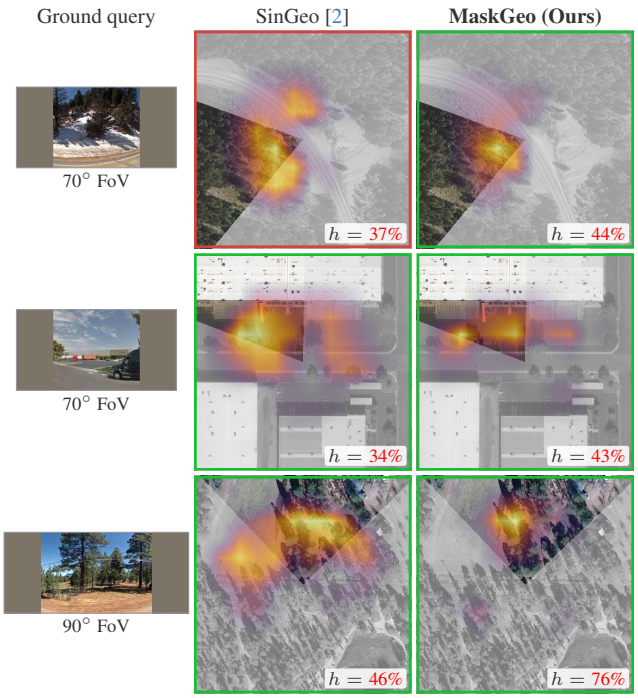


Figure 5. Comparing aerial activations for MaskGeo and SinGeo trained for 64px padding. The colored region of the aerial image is the region visible to the ground image.

5. Conclusion

In this work we present MaskGeo, which introduces ground and aerial view masking for robust CVGL.

To the best of our knowledge, this is the first work in cross-view geo-localization to mask both ground and aerial views under a curriculum and to rank the masked views between positives and negatives.

We also emphasize that contrastive-only losses gated by the overlap pseudo label strengthen the robustness to unknown orientation and varying field of view. We show that the use of semi-positives helps the model generalize well and learn a better representation for limited-FoV CVGL. MaskGeo also explores the use of padding for limited-FoV imagery, which enhances the applicability of MaskGeo to transformer backbones and we show that it can be accomplished without hindering the practicality at inference time.

MaskGeo relies on training-image metadata to generate the overlap pseudo labels, which is a limitation. Future work can rank several masked views by their continuous overlap with the anchor, for which Rank-N-Contrast is a natural extension.

The code will be made publicly available to facilitate further research in this area.

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